👩🏾‍🏫 Staff Guide · Biology A · Unit 5 Breathing, Exercise & Energy (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in Ms. Dunham's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: "Unit 3 Diabetes: A2 Exercise Lab, A3 Combustion, A4 Yeast Feast, A5 Combustion & Exercise, A5 Respiration (ATP video + ppt_ATP), Heart/Lungs, A7 Putting It All Together, Metabolic pathways")	**Level 1 piece**
A2 Exercise Lab	Exercise lab packet + notes part A
A3 Combustion + A3.5 Combustion & Exercise (Mallow Out, Liquid on Fire)	Combustion demo packet + notes part B (grouped)
A4 Yeast Feast	Yeast feast packet + notes part C
A5 Respiration (ATP video, yeast balloons) + Heart/Lungs + Metabolic pathways	Notes part D + slides (grouped as one lesson)
Snail/Elodea day 1 + final	Elodea & Snails packet in Unit 2 (her demo sheet was found)
Plant sprout lab	🚧 Under construction (title only)
Class slides	Deck: 19 slides, Words to Know first, "point to the picture" checks, recap
Her worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	10 picture cards (cut apart)
—	2 games on the class site: matching + step-by-step order

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Picture anchors: 🔥 cells burn food, 🪙 ATP = energy coins, 🟡 BTB yellow = CO2
- Level 1: no formulas. Level 2: word equation only

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the "look for" part
- Cover one choice to make it 2

4. What to grade

- Quiz stands in for this part of the paper Diabetes test
- **Notes**: same points as her worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver, Room 113 (SLE): changes and IEP questions.
- Facts we could not confirm in her files say "check with Ms. Dunham".

🚧 Under construction (check with Ms. Dunham)

- **Plant sprout lab**: Only a 4-question data quiz; no lab directions.
- **A5 Respiration amounts**: Her yeast-balloon lab leaves the grams of yeast and sugar blank (set by her).

💗 Sensitive topic note

Her A3 Combustion compares a weight-loss contestant (Rachel, Biggest Loser) to a marshmallow and uses before/after photos. We left the person and the photos out: our version talks only about the marshmallow and "our cells." Exercise: follow each student's IEP/504 limits.

6. Photo credits (open licenses)

Human Lungs (NIH BioArt 231 - 630892).png — NIAID / Ryan Kissinger (NIH BioArt), Public domain, Wikimedia Commons · Hand Holdng Fire Toasted Marshmallow (3310040440).jpg — Pink Sherbet Photography from USA, CC BY 2.0, Wikimedia Commons · Running girl at the sports day of Baozhong Junior High School 2015-11-13.jpg — 褒忠國中 雲端網, CC BY 2.0, Wikimedia Commons

👩‍🏫 Staff Guide · Combustion Demo: Mallow Out (Level 1)

Biology A · Unit 5 · goes with the class “A3 Combustion / A3.5 Combustion & Exercise” (5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Observer card**: A3 Combustion + A3.5 Part 2 “Mallow Out” and Part 3 “Liquid on Fire” (ethanol): teacher demo, our students watch and record. 5 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Smaller or bigger?”
- “Where did it go?”

3. Modifications

- Watch-only role
- Circle answers

4. What to grade (5 pts)

- Same 5 points: circles + job
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

💗 Sensitive topic note

Her A3 asks what “Rachel then/now” (a Biggest Loser contestant) and a marshmallow have in common, with before/after photos. We removed the person and the photos. Talk only about the marshmallow and cells.

6. Photo credits (open licenses)

Hand Holdng Fire Toasted Marshmallow (3310040440).jpg — Pink Sherbet Photography from USA, CC BY 2.0, Wikimedia Commons

👩‍🏫 Staff Guide · Exercise Lab (Level 1)

Biology A · Unit 5 · goes with the class “A2 Exercise Lab” (5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A2 Exercise Lab: breathing rate, heart rate, and CO2 (BTB) at rest and after 2 minutes of exercise. 6 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Faster or slower?”
- “Say yellow when you see it.”
- “Why does your heart speed up?”

3. Modifications

- Any OK'd movement counts
- Staff counts with the student; circle faster/slower
- Staff may do the BTB blowing; student is color checker

4. What to grade (5 pts)

- Same 5 points: circles + job
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

💗 Sensitive topic note

Exercise and breathing: check each student's IEP/504 and health plan (asthma, heart, mobility) before the lab. Any movement counts.

6. Photo credits (open licenses)

Running girl at the sports day of Baozhong Junior High School 2015-11-13.jpg — 褒忠國中 雲端網, CC BY 2.0, Wikimedia Commons

👩🏾‍🏫 Staff Guide · Yeast Feast (Level 1)

Biology A · Unit 5 · goes with the class “A4 Yeast Feast” (5 pts). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Lab card**: A3.4 Yeast Feast (2025): 8 mL food + warm yeast in a tube; measure the CO2 gas that collects. 5 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 2 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- “Which tube has the biggest bubble?”
- “Is yeast alive?”

3. Modifications

- Circle the tube with the most gas
- Staff helps read the tubes

4. What to grade (5 pts)

- Same 5 points: circle + job
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver, Room 113 (SLE).

🚧 Under construction (check with Ms. Dunham)

- **This packet**: Her sheet leaves the 4th food as “Your choice.” Please tell us which choice foods are allowed so we can do an allergy check and add the picture.

🫁 Breathing, Exercise & Energy Notes

Biology A · Unit 5 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

🫁 lungs _____

❤️ heart _____

🪙 energy _____

💨 air _____

🖼️ B. Fill in the word. Use the pictures.

5. 🍬 Cells burn _____ for energy.

6. 💨 We breathe in _____.

7. 🩸 The heart pumps _____.

8. ⬆️ Exercise makes my heart go _____.

Word bank: 🩸 **blood** | ⬆️ **faster** | 💨 **oxygen** | 🍬 **sugar**

⭕ C. Circle the right picture.

9. We breathe out…



Sugar



Carbon dioxide

10. Energy coins are…



Butter



ATP

11. BTB with CO2 turns…



Yellow



Blue

🔗 D. Draw a line from the picture to the word.









heart

yeast

burn

lungs

📖 Breathing, Exercise & Energy: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Cellular respiration

cells burn food for energy



ATP

energy coins for cells



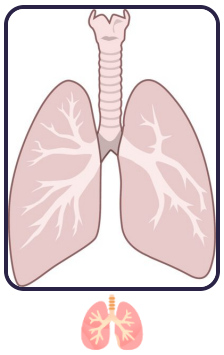
Oxygen

the air we breathe in



Carbon dioxide

the air we breathe out



Lungs

breathe air in and out

Heart

pumps blood

Combustion

burning

Yeast

tiny living thing that eats sugar

Exercise

moving my body hard

BTB

blue liquid turns yellow with CO_2

🔥 Combustion Demo: Mallow Out

Biology A · Unit 5 · Level 1



Name: _____

A3 Combustion + A3.5 Part 2 “Mallow Out” and Part 3 “Liquid on Fire” (ethanol): teacher demo, our students watch and record.

🧺 Demo (teacher does it): get it. Check ☐.

- ☐ 🍡 **1** marshmallow on a skewer
- ☐ ⚖️ **1** scale
- ☐ 🔥 **1** lighter (teacher only)
- ☐ 🕯️ **1** tea-light cup with ethanol (teacher only)

🌱 Do one box. Check ☐ it.

1. ⚖️ **Weigh** the marshmallow. ☐
2. 🔥 Teacher **burns** it. I watch. ☐
3. ⌛ **Wait** for it to cool. ☐
4. ⚖️ **Weigh** it again: smaller! ☐
5. 💨 Where did it go? **Into the air**. ☐

📋 Circle.

After burning, the marshmallow is: ⬇️ smaller ⬆️ bigger

The missing part went: 💨 into the air 🗑️ in the trash

🏃 Exercise Lab

Biology A · Unit 5 · Level 1



Name: _____

A2 Exercise Lab: breathing rate, heart rate, and CO2 (BTB) at rest and after 2 minutes of exercise.

🧺 Lab stuff: get it. Check ☐.

- ☐ 🥛 **20 mL** water in a 150 mL beaker
- ☐ 🔵 **8 drops** BTB (staff adds)
- ☐ 🥤 **1** straw (blow OUT only)
- ☐ ⏱️ **1** timer

🌱 Do one box. Check ☐ it.

1. 🪑 **Rest**. Count breaths (staff helps). ☐
2. ❤️ Feel my **heartbeat**. ☐
3. 🏃 **Move** for 2 minutes. ☐
4. ❤️ Count **again**. Faster? ☐
5. 🟡 Watch BTB turn **yellow**. ☐
6. 📝 Circle faster or slower. ☐

📋 **Circle.**

After moving, breathing: ⬆️ faster ⬇️ slower

After moving, heart: ⬆️ faster ⬇️ slower

BTB turned: 🟡 yellow 🔵 stayed blue

🍞 Yeast Feast

Biology A · Unit 5 · Level 1



Name: _____

A3.4 Yeast Feast (2025): 8 mL food + warm yeast in a tube; measure the CO2 gas that collects.

🧺 Foods to test (her list): get it. Check ☐.

- ☐ 🍬 **8 mL** glucose solution
- ☐ 🍞 **8 mL** starch solution
- ☐ 💧 **8 mL** water (control)
- ☐ ❓ **8 mL** your choice food (staff OK)

🌱 Do one box. Check ☐ it.

1. 🧪 Put **food** in the tube. ☐
2. 🍞 Add **warm yeast**. ☐
3. 👍 Cap it. Flip it in warm water. ☐
4. 🫧 Watch the **bubbles** grow. ☐
5. 📏 Which tube has the **most** gas? ☐

🫧 Circle the tube with the most bubbles.

🍬 🍞 💧 ❓

🚧 **Under construction:** Her sheet leaves the 4th food as “Your choice.” Please tell us which choice foods are allowed so we can do an allergy check and add the picture. *Check with Ms. Dunham.*

🔥 Combustion Demo: Mallow Out · Get Ready

Biology A · Unit 5 · Level 1



Name: _____

👥 My job. Circle it.

⚖️	👀	📝	📷
Scale reader I read the scale.	**Watcher** I watch.	**Recorder** I write.	**Photo taker** I take pictures.

🧰 Get the tools. Check ☐.

- ☐ 🥽 Goggles
- ☐ ⚖️ Scale
- ☐ 📋 Observer card
- ☐ 🧯 Fire safety

⚠️ Be safe!

- 🔥 Stay back from fire.
- 🥽 Goggles on.
- 🍡 Do not eat it.
- ⏳ Wait. It is hot.

✅ Before we start

☐ Goggles on ☐ Behind the line ☐ Allergy/diet check done

🏃 Exercise Lab · Get Ready

Biology A · Unit 5 · Level 1



Name: _____

👥 My job. Circle it.

⏱️	❤️	🟡	📝
Timer	**Counter**	**Color checker**	**Recorder**
I watch the clock.	**I count.**	**I see yellow.**	**I write it.**

🧰 Get the tools. Check ☐.

- ☐ ⏱️ Timer
- ☐ 🥤 Straw
- ☐ 🥽 Goggles
- ☐ 📋 Data table

⚠️ Be safe!

- 🥤 Blow out only!
- 🥽 Goggles on.
- 🏃 Move the way that is OK for me.
- 💧 Tell staff if I feel sick.

✅ Before we start

☐ Goggles on ☐ My straw ☐ My exercise is OK'd by staff

🍞 Yeast Feast · Get Ready

Biology A · Unit 5 · Level 1



Name: _____

👥 My job. Circle it.

 **Pourer** I pour.	 **Yeast adder** I add yeast.	 **Reader** I read the bubbles.	 **Recorder** I write.

🧰 Get the tools. Check ☐.

- ☐  4 tubes
- ☐  Beakers
- ☐  Thermometer
- ☐  Timer

⚠️ Be safe!

- 🥽 Goggles on.
- 🚫 No tasting.
- 🍬 Allergy check first.
- 🧼 Wash hands.

✅ Before we start

☐ Goggles on ☐ Allergy check done ☐ 4 tubes labeled

🔥 Combustion Demo: Mallow Out · My Check

Biology A · Unit 5 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🍡 Before
- ☐ 🔥 After

✅ I did it! Check ☐.

- ☐ 👀 I watched safely
- ☐ ⚖️ I saw the mass change
- ☐ 🤝 I did my job

⭕ Circle one face.

🔥 **Was I safe?**	😀	🙂	😕
🤝 **Did I do my job?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

🏃 Exercise Lab · My Check

Biology A · Unit 5 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🟡 Yellow BTB
- ☐ 📋 My circles

✅ I did it! Check ☐.

- ☐ ❤️ I counted
- ☐ 🟡 I saw the color
- ☐ 🥤 I blew out only
- ☐ 🤝 I did my job

⭕ Circle one face.

🥤 **Was I safe?**	😀	🙂	😕
🤝 **Did I do my job?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

🚧 Plant Sprout Lab

Biology A · Unit 5 · Level 1 · Under construction

Name: _____

🚧 **Under construction:** Only a 4-question data quiz is in the export (hypothesis supported? photo of tubes, plants do photosynthesis and respiration). No lab directions. *Check with Ms. Dunham.*

What we have so far

Quiz Q3 key: animals do respiration; plants do photosynthesis and respiration.

Staff: until this is ready

Have the student do the matching Level 1 unit game or vocabulary cards for this lesson, and give the same points for effort. Mr. Driver, Room 113.

🍞 Yeast Feast · My Check

Biology A · Unit 5 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 🧪 My 4 tubes

✅ I did it! Check ☐.

- ☐ 🫧 I saw the bubbles
- ☐ 🧠 I picked the most gas
- ☐ 🤝 I did my job

⭕ Circle one face.

🫧 **Did I see the bubbles?**	😀	🙂	😕
🤝 **Did I do my job?**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Breathing, Exercise & Energy Quiz

Biology A · Unit 5 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🔥 Cells burn food to get…

- 💧 Only water
- ⚡ Energy

2. 🪙 Energy coins in cells are called…

-  ATP
-  Bread

3. 🏃 After exercise, breathing is…

- ⬆️ Faster
- ⬇️ Slower

4. 🟡 BTB turns yellow when we add…

-  Ice
-  Carbon dioxide

5. 💨 Cells need sugar and…

-  Rocks
-  Oxygen

6. 🍞 Yeast eating sugar makes…

-  Gas bubbles
-  Ice

7. 🔥 A burning marshmallow gets…

-  Bigger
-  Smaller

8. ❤️ The heart pumps…

-  Blood
-  Air

🫁 Breathing, Exercise & Energy Notes · KEY

Biology A · Unit 5 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

🫁 lungs _____

❤️ heart _____

🪙 energy _____

💨 air _____

🖼️ B. Fill in the word. Use the pictures.

5. 🍬 Cells burn **sugar** for energy.

6. 💨 We breathe in **oxygen**.

7. 🩸 The heart pumps **blood**.

8. ⬆️ Exercise makes my heart go **faster**.

Word bank: 🩸 **blood** | ⬆️ **faster** | 💨 **oxygen** | 🍬 **sugar**

⭕ C. Circle the right picture.

9. We breathe out…

- Sugar
- Carbon dioxide

10. Energy coins are…

- Butter
- ATP

11. BTB with CO2 turns…

- Yellow
- Blue

🔗 D. Draw a line from the picture to the word.

Key: 🫁 → lungs · ❤️ → heart · 🍞 → yeast · 🔥 → burn

🫁	heart
❤️	yeast
🍞	burn
🔥	lungs

🔑 Key written by our team from Ms. Dunham's materials, please check.

✅ Breathing, Exercise & Energy Quiz · KEY

Biology A · Unit 5 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. 🔥 Cells burn food to get… A3.10 review

- 💧 Only water
- ✅ ⚡ Energy

2. 🪙 Energy coins in cells are called… A5 Respiration

- ✅ 🪙 ATP
- 🍞 Bread

3. 🏃 After exercise, breathing is… A2 Exercise Lab

- ✅ ⬆️ Faster
- ⬇️ Slower

4. 🟡 BTB turns yellow when we add… A2 Exercise Lab

- 🧊 Ice
- ✅ 😮‍💨 Carbon dioxide

5. 💨 Cells need sugar and… A3.10 review

- 🪨 Rocks
- ✅ 💨 Oxygen

6. 🍞 Yeast eating sugar makes… A3.10 review

- **Gas bubbles** 
- **Ice** 

7. 🔥 A burning marshmallow gets… A3 Combustion

- **Bigger** 
- **Smaller** 

8. ❤️ The heart pumps… Heart/Lungs

- **Blood** 
- **Air** 

Answers

1-B (Energy) · 2-A (ATP) · 3-A (Faster) · 4-B (Carbon dioxide) · 5-B (Oxygen) · 6-A (Gas bubbles) · 7-B (Smaller) · 8-A (Blood)

Level 1 version of the gen ed “A3.10 Diabetes Review (A3.4-A3.5 parts) + A2, A3, A5 sheets”. Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from her A3.10 review and the A2/A3/A5 sheets, please check.*